Solution of MGMT 670 Homework 3

Due 11:59 pm EST on Oct 6. Submit via Brightspace.

1. We are curious about the average number of e-mails that Purdue's MS-BAIM students reply to daily. A sample of 40 students was randomly selected, and the number of e-mails they replied to daily are listed below:

   $\{5, 5, 1, 3, 1, 17, 5, 0, 4, 3, 0, 1, 13, 3, 12, 2, 1, 13, 5, 14, 9, 10, 5, 2, 1, 8, 2, 26, 4, 0, 6, 6, 7, 12, 8, 22, 5, 7, 24, 2\}$

   The following descriptive statistics for the data are provided:

Variable	N	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
Emails	40	6.85	1.04	6.56	0.00	2.00	5.00	9.75	26.00

   (I) Testing the Mean Number of E-mails:

   Dave Foster wants to test if the average number of e-mails that Purdue's MS-BAIM students reply to daily is **more than 5**.

   (a) State the appropriate null and alternative hypotheses.

   (b) Compute the value of the test-statistic.

   (c) Using $\alpha = 0.05$, would Dave Foster conclude that the mean number of e-mails that MS-BAIM students reply to daily is more than 5? Explain.

   (II) Testing the Proportion of MS-BAIM Students Replying to More than 4 E-mails:

   Dave Foster also wants to test whether more than 50% of MS-BAIM students reply to **more than 4 e-mails daily**. Use the sample in this problem to construct a hypothesis test.

   (d) State the appropriate null and alternative hypotheses.

   (e) Compute the value of the test-statistic.

   (f) Using $\alpha = 0.10$, would Dave Foster conclude that more than 50% of MS-BAIM students reply to more than 4 e-mails daily? Explain.

   Solution:

   (I) Testing the Mean Number of E-mails:

   (a) Hypotheses We are testing whether the mean number of e-mails replied to by MS-BAIM students is greater than 5.

   - **Null Hypothesis** (H_0): $\mu \leq 5$, where μ is the true mean number of e-mails replied to daily by MS-BAIM students. It's also OK to say (H_0): $\mu = 5$
   - **Alternative Hypothesis** (H_1): $\mu > 5$.

(b) Test Statistic We use the one-sample t-test for the mean. The test statistic is given by:

$$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$$

Where:

- $\bar{x} = 6.85$ (sample mean),
- $\mu_0 = 5$ (hypothesized mean),
- $s = 6.56$ (sample standard deviation),
- $n = 40$ (sample size).

Substitute the values:

$$t = \frac{6.85 - 5}{\frac{6.56}{\sqrt{40}}} = \frac{1.85}{1.04} \approx 1.78.$$

(c) Conclusion For $\alpha = 0.05$, we check the critical value from the t-distribution with $n - 1 = 39$ degrees of freedom. The critical t-value for a one-tailed test is approximately $t_{0.05,39} = 1.685$.

Since $t = 1.78 > t_{0.05,39} = 1.685$, we reject the null hypothesis. There is sufficient evidence at the 5% significance level to conclude that the mean number of e-mails replied to daily is greater than 5.

(II) Testing the Proportion of MS-BAIM Students Replying to More than 4 E-mails:

(d) Hypotheses We are testing whether more than 50% of MS-BAIM students reply to more than 4 e-mails daily.

- **Null Hypothesis** (H_0): $p \leq 0.5$, where p is the proportion of MS-BAIM students who reply to more than 4 e-mails daily.
- **Alternative Hypothesis** (H_1): $p > 0.5$.

(e) Test Statistic We use a one-sample z-test for the proportion. The sample contains 23 students who replied to more than 4 emails, out of a total sample size of 40 students. So, the sample proportion $\hat{p}$ is:

$$\hat{p} = \frac{23}{40} = 0.575$$

We use a one-sample z-test for proportions. The test statistic is calculated using the formula:

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

where:

- $\hat{p} = 0.575$ (sample proportion),
- $p_0 = 0.50$ (hypothesized proportion),
- $n = 40$ (sample size).

Substituting the values:

$$z = \frac{0.575 - 0.50}{\sqrt{\frac{0.50(1-0.50)}{40}}} = \frac{0.075}{0.0791} \approx 0.949$$

(f) Conclusion

Using $\alpha = 0.10$, the critical value for a one-tailed z-test at $\alpha = 0.10$ is approximately $z_{0.10} = 1.28$.

Since the calculated z-statistic (0.949) is less than the critical value (1.28), we fail to reject the null hypothesis. There is insufficient evidence at the 10% significance level to conclude that more than 50% of MS-BAIM students reply to more than 4 e-mails daily.

2. The office of records at a university has stated that the proportion of incoming female students who major in business has increased. A sample of female students taken several years ago is compared with a sample of female students this year. Results are summarized below. Has the proportion increased significantly? Test at $\alpha = 0.10$.

	Sample Size	Numbers Majoring in Business
Previous Sample	250	50
Present Sample	300	90

Solution

(a) **Hypotheses:** Let p_1 be the proportion of female students majoring in business previously, and p_2 be the proportion currently.

$$H_0 : p_2 - p_1 \leq 0 \quad \text{(no increase)}, \qquad H_a : p_2 - p_1 > 0 \quad \text{(increase)}$$

(b) **Sample proportions:**

$$\bar{p}_1 = \frac{50}{250} = 0.20, \quad \bar{p}_2 = \frac{90}{300} = 0.30$$

(c) **Pooled proportion under H_0:**

$$\bar{p} = \frac{50 + 90}{250 + 300} = \frac{140}{550} \approx 0.2545$$

(d) **Standard error:**

$$SE = \sqrt{\bar{p}(1 - \bar{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)} = \sqrt{0.2545 \cdot 0.7455 \left(\frac{1}{250} + \frac{1}{300} \right)} \approx 0.037$$

(e) **Test statistic:**

$$z = \frac{\bar{p}_2 - \bar{p}_1}{SE} = \frac{0.30 - 0.20}{0.037} \approx 2.70$$

(f) **Critical value and decision:** For $\alpha = 0.10$ (one-tailed), $z_{0.10} \approx 1.28$. Since $z_{\text{calc}} = 2.70 > 1.28$, we reject H_0.

(g) **Conclusion:** At the 10% significance level, there is strong evidence that the proportion of incoming female students majoring in business has **significantly increased**.

3. A newspaper in a large city, whose daily circulation is currently 190,000, is thinking about issuing a Sunday edition. The business development team calculated that in order for the Sunday edition to be profitable, the circulation needs to be at least 200,000.

   The development team collected data from 35 other newspapers in big cities in order to forecast the circulation of their potential Sunday edition. The data is provided below (the numbers are in thousands):

Paper	Sunday Circulation	Daily Circulation
Des Moines Register	344.522	206.203
Philadelphia Inquirer	982.663	515.523
Tampa Tribune	408.342	321.626
New York Times	1762.015	1209.224
New York News	983.239	781.796
Sacramento Bee	338.355	273.843
Los Angeles Times	1531.526	1164.387
Boston Globe	798.297	516.981
Cincinnati Enquirer	348.743	198.832
Orange Co. Register	407.760	354.842
Miami Herald	553.479	444.580
Chicago Tribune	1133.249	733.775
Detroit News	1215.149	481.765
Houston Chronicle	620.752	449.755
Kansas City Star	423.304	288.571
Omaha World Herald	284.610	223.748
Denver Post	417.778	252.623
St. Louis Post-Dispatch	585.681	391.286
Portland Oregonian	440.923	337.671
Washington Post	1165.567	838.901
Long Island Newsday	960.307	825.512
San Francisco Chronicle	704.322	570.364
Chicago Sun Times	559.093	537.780
Minneapolis Star Tribune	685.974	412.871
Baltimore Sun	488.506	391.951
Pittsburgh Press	557.000	220.464
Rocky Mountain News	432.502	374.009
Boston Herald	235.083	355.627
New Orleans Times-Picayune	324.240	272.279
Charlotte Observer	299.450	238.554
Hartford Courant	323.084	231.177
Rochester Democratic and Chronicle	262.048	133.238
St. Paul Pioneer Press	267.781	201.860
Providence Journal-Bulletin	268.059	197.119
L.A. Daily News	202.613	185.735

Below is the Minitab output for Regression Analysis: Sunday versus Daily

Analysis of Variance

Source	DF	SS	MS	F-Value	P-Value
Regression	1	4370975	4370975	211.19	0.000
Error	33	683001	20697		
Total	34	5053976			

Model Summary

S	R-sq	R-sq(adj)
143.865	86.5%	86.1%

Coefficients

Term	Coef	SE Coef	T-Value	P-Value
Constant	24.76	46.99	0.53	0.602
Daily	1.3512	0.09298	14.53	0.000

Regression Equation

$$\text{Sunday} = 24.8 + 1.3512 \times \text{Daily}$$

Prediction for Sunday

$$\text{Sunday} = 24.8 + 1.3512 \times 190$$

Fit	SE Fit	90% CI	90% PI
281.486	33.1564	$(225.374, 337.599)$	$(31.6333, 531.340)$

Questions

(a) What's the correlation coefficient between the circulation of daily and Sunday editions? How strong is the relationship?

(b) One of the team members claims that increasing 1000 in daily circulation would increase more than 1300 in Sunday edition. Does this data provide enough evidence (5% level of significance) to support that claim?

(c) Give 95% confidence interval for the slope of the regression.

(d) Explain the difference between the 90% confidence interval for the mean Sunday circulation and the 90% prediction interval for a single newspaper's Sunday circulation. Why is the prediction interval wider?

(a) **What's the correlation coefficient between the circulation of daily and Sunday editions? How strong?**

The coefficient of determination, R^2, is given as 86.5%. The correlation coefficient, r, is the square root of R^2, so:

$$r = \sqrt{0.865} = 0.930.$$

Therefore, the correlation coefficient is approximately 0.930. Since the slope is positive, $r \approx 0.930$, indicating a very strong positive relationship between daily and Sunday circulation.

(b) **Test if each additional 1000 in daily circulation increases Sunday circulation by more than 1300 copies:**
Null and alternative hypotheses:

$$H_0 : \beta_1 = 1.3, \quad H_a : \beta_1 > 1.3$$

From the regression output: $\hat{\beta}_1 = 1.3512$, $SE_{\hat{\beta}_1} = 0.09298$.
Test statistic:

$$t = \frac{\hat{\beta}_1 - 1.3}{SE_{\hat{\beta}_1}} = \frac{1.3512 - 1.3}{0.09298} \approx 0.552$$

Degrees of freedom: $df = n - 2 = 33$. Critical t-value at $\alpha = 0.05$ (one-tailed): $t_{0.05,33} \approx 1.692$. Since $t_{\text{calc}} = 0.552 < 1.692$, we fail to reject H_0.
Conclusion: There is not enough evidence to conclude that each additional 1000 in daily circulation increases Sunday circulation by more than 1300 copies.

(c) **Give 95% confidence interval for the slope of the regression.**

$$\hat{\beta}_1 \pm t_{0.025,33} \cdot SE_{\hat{\beta}_1}, \quad t_{0.025,33} \approx 2.034$$

$$1.3512 \pm 2.034 \times 0.09298 \approx 1.3512 \pm 0.189$$

$$95\% \text{ CI: } (1.162, 1.540)$$

Interpretation: We are 95% confident that for each 1000 increase in daily circulation, Sunday circulation increases between 1162 and 1540 copies.

(d) **Difference between 90% confidence interval and 90% prediction interval:** - The 90% confidence interval estimates the *mean* Sunday circulation for newspapers with a given daily circulation. - The 90% prediction interval estimates the range of Sunday circulation for a *single* newspaper. - Prediction interval is wider because it accounts for both the uncertainty in estimating the mean and the random variability of individual newspapers.